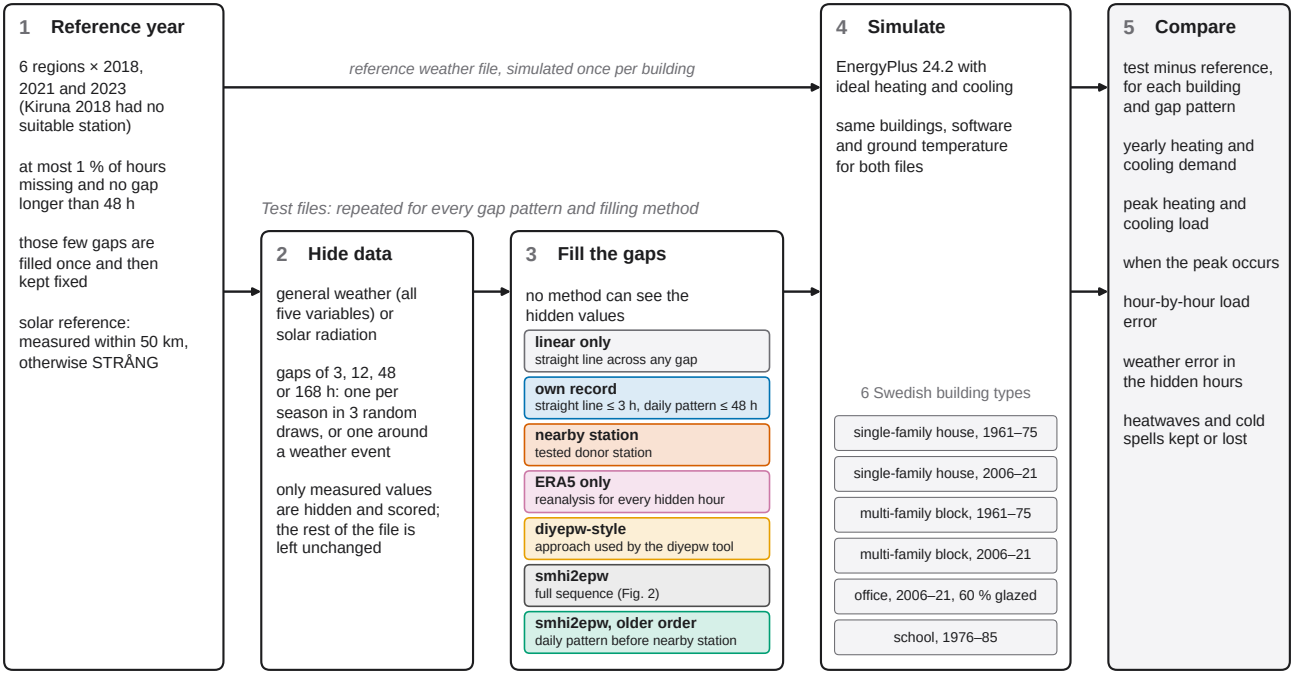
1 Reference year
6 regions × 2018, 2021 and 2023 (Kiruna 2018 had no suitable station)
at most 1 % of hours missing and no gap longer than 48 h
those few gaps are filled once and then kept fixed
solar reference: measured within 50 km, otherwise STRÅNG
reference weather file, simulated once per building
Test files: repeated for every gap pattern and filling method
2 Hide data
general weather (all five variables) or solar radiation
gaps of 3, 12, 48 or 168 h: one per season in 3 random draws, or one around a weather event
only measured values are hidden and scored; the rest of the file is left unchanged
3 Fill the gaps
no method can see the hidden values
linear only
straight line across any gap
own record
straight line ≤ 3 h, daily pattern ≤ 48 h
nearby station
tested donor station
ERA5 only
reanalysis for every hidden hour
diyepw-style
approach used by the diyepw tool
smhi2epw
full sequence (Fig. 2)
smhi2epw, older order
daily pattern before nearby station
4 Simulate
EnergyPlus 24.2 with ideal heating and cooling
same buildings, software and ground temperature for both files
6 Swedish building types
single-family house, 1961–75
single-family house, 2006–21
multi-family block, 1961–75
multi-family block, 2006–21
office, 2006–21, 60 % glazed
school, 1976–85
5 Compare
test minus reference, for each building and gap pattern
yearly heating and cooling demand
peak heating and cooling load
when the peak occurs
hour-by-hour load error
weather error in the hidden hours
heatwaves and cold spells kept or lost